Quality Information Document

Ocean Monitoring Indicator (OMI) of the Copernicus Marine Service

EXTREME SL

Sea level

Issue: 2.1

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
	22/07/2019	All	Creation of the document	Marta de Alfonso, Alejandro Gallardo, Fernando Manzano, Begoña Pérez Gómez	
1.0	02/10/2019	All	Adapt the document to the new OMI structure and naming	Marta de Alfonso	
2.0	02/09/2022		New Product naming and new template, modified according to new sea level REP product input	Marta de Alfonso, Begoña Pérez Gómez	Stéphane Tarot
2.1	07/05/2025		Estimation of uncertainties with standard deviation of percentiles	Marta de Alfonso, Begoña Pérez Gómez, Alex Gallardo	Stéphane Tarot

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I INTRODUCTION

I.1 Product name

OMI_EXTREME_SL_BALTIC_slev_mean_and_anomaly_obs

OMI_EXTREME_SL_NORTHWESTSHELF_slev_mean_and_anomaly_obs

OMI_EXTREME_SL_IBI_slev_mean_and_anomaly_obs

OMI_EXTREME_SL_MEDSEA_slev_mean_and_anomaly_obs

I.2 Indicator evaluation method

The EXTREME SL indicators are based on the computation of the 99th and the 1st percentiles (that represent approximately the 1% highest and lowest values, respectively) of hourly sea levels derived from in situ sea level observations, following the same approach applied by Woodworth and Blackman (2004) to extreme sea levels. This method determines changes in the frequency distribution of the measured variables.

All the data used for the computation of these indicators have been downloaded from Copernicus Marine Service In Situ Thematic Assembly Centre (In Situ TAC) products. The analysis is performed for the following variable: coastal sea level (slev, total sea level at each tide gauge, including the tide) measured by in situ tide gauges (hourly values).

The periods selected are based on our analysis of the data available in the different In Situ TAC regions that fulfilled specific criteria of data completeness and quality (Pérez Gómez et al 2016, Pérez Gómez et al 2018). Tide gauges were selected according to consolidated and reprocessed time series of hourly sea level data in Copernicus Marine Service for the period 1993 to the target year, product INSITU_GLO_PHY_SSH_DISCRETE_MY_013_053. Within these periods, a minimum number of 15 years with completeness index larger than 70% is required for a station to be included. These data were then used to compute annually the 99th and 1st percentile levels at each station.

More details can be found in the Copernicus Marine Service Ocean State Report (Pérez Gómez et al., 2016; Pérez Gómez et al., 2018).

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I.3 Products used as input for the computations

Product name (and Type)	Documentation
INSITU_GLO_PHY_SSH_DISCRETE_MY_013_053	QUID: https://doi.org/10.13155/99217 PUM: https://doi.org/10.13155/97083
INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030 INSITU_BAL_PHYBGCWAV_DISCRETE_MYNRT_013_032 INSITU_NWS_PHYBGCWAV_DISCRETE_MYNRT_013_036 INSITU_IBI_PHYBGCWAV_DISCRETE_MYNRT_013_033 INSITU_MED_PHYBGCWAV_DISCRETE_MYNRT_013_035	QUID: https://doi.org/10.13155/75807 PUM: https://doi.org/10.13155/43494

II SCIENTIFIC VALIDATION

The quality validation of the EXTREME SL indicators is based on the validation of the original in situ data used to compute the percentiles.

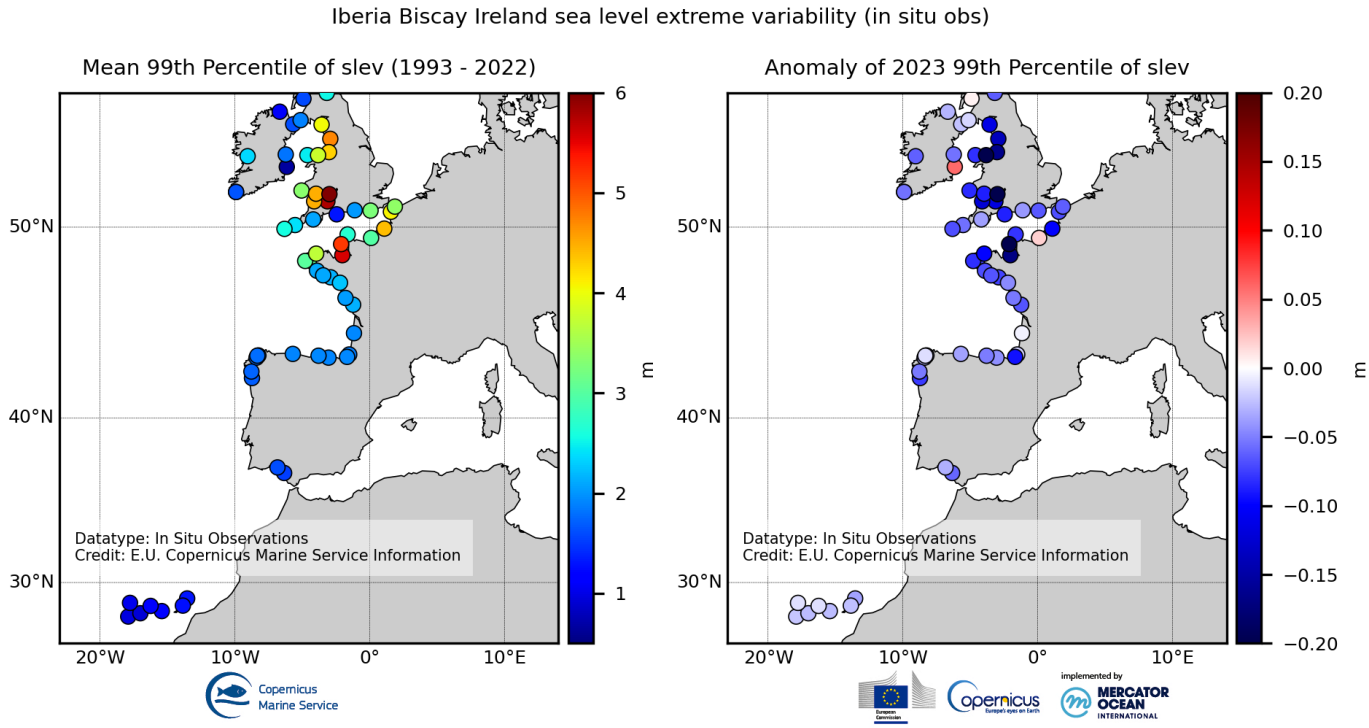
Whenever available, data have been selected from the In Situ TAC REProcessed products (REP), in this case product INSITU_GLO_PHY_SSH_DISCRETE_MY_013_053, where data have passed a validation and assessment process. To know more about these processes, the Quality Information Document of the products can be consulted in https://doi.org/10.13155/99217.

When needed, the information has been completed with the In Situ TAC Near Real Time products (NRT) for recent data to perform the analysis. Accuracy numbers about the original data and the quality procedures applied to the NRT products can be found in the Quality Information Document of NRT In Situ TAC Products (https://doi.org/10.13155/75807). Complete information about the quality procedures applied to In Situ TAC NRT data can be found in the Copernic Marine In Situ TAC Quality Control Procedures (https://doi.org/10.13155/106219).

For these NRT data, a visual inspection by scientists has been performed to assure that no spurious value interferes and deviate the extremes results.

In a normal year with no data gaps, we have 8760 hourly values (6132 values for the minimum data coverage of 70%). For these numbers, the 99th and 1st percentiles represent the 88 highest and lowest hourly values, respectively (or 61 for the 70% minimum data coverage). This makes this extremes study relatively immune to individual data measurement errors instead of annual maximum and minimum values.

Apart from the robustness provided using percentiles for the extremes study, we check the consistency by performing an analysis of the spatial distribution, starting with the calculation of the percentiles for every location and for every year. The annual percentiles of slev are referred to local mean sea level at each tide gauge (subtracting the annual 50th percentile). In the analysis (corresponding to the products: OMI_EXTREME_SL_[region]_ slev_mean_and_anomaly_obs), the temporal mean percentile is calculated with the whole series until the year previous to the target year and its spatial distribution is shown together with the anomaly of the target year. This analysis allows to check the geographical coherence of the OMI (see Figure below).



Iberia Biscay Ireland sea level 99th percentile. Left: Mean of annual values (1993 - 2022). Right: Anomalies of 2023 with respect to the mean.

More details can be found in the Copernicus Marine Service Ocean State Report (Pérez Gómez et al., 2016; Pérez Gómez et al., 2018).

Uncertainty

Uncertainty refers to the error associated with estimating a parameter, such as the mean, and arises due to inherent variability within a sample. This variability is quantified by the standard deviation, a statistical measure that characterizes the dispersion of data points around the mean, thereby providing an indication of the uncertainty involved in the estimation.

The standard deviation included in this indicator defines the threshold for statistical significance, with anomalies considered significant if they deviate substantially from it.

III REFERENCES

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